

## CLASS TOPIC

# Inference in regression; correlation vs causation

## What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

**W13****Statistics**

Clean formulas · aligned layout ·  
corrected terminology



## CLASS TOPIC

# Inference in regression; correlation vs causation

## Economic question first

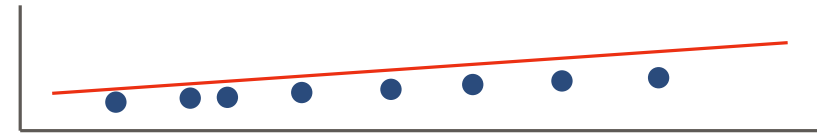
- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

**FORMULA**

$$H_0: \beta_1 = 0$$

$$t = (b_1 - 0) / SE(b_1)$$

$$CI: b_1 \pm t^*SE(b_1)$$



## CLASS TOPIC

# Inference in regression; correlation vs causation

## Key definitions

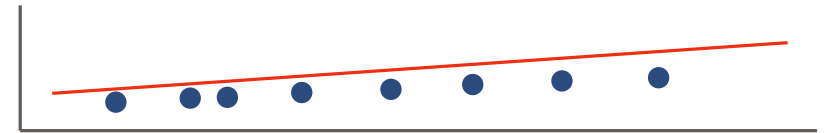
- Population: full group of interest.
- Sample: observed part of the population.
- Statistic: number computed from a sample;  
parameter: number for the population.

**FORMULA**

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## Main formula box

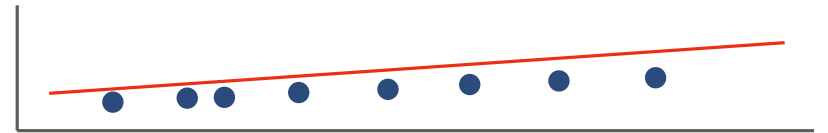
- All formulas are kept in one merged formula area on this slide.
- Read every symbol before calculating.
- Use parentheses carefully.

**FORMULA 1**

$$H_0: \beta_1 = 0$$

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## Mini example

- Write the data clearly.
- Substitute values step by step.
- Round only at the final reporting stage.

**01** Data: 4, 7, 8, 9, 12

**02** Step 1: compute the needed statistic.

**03** Step 2: write the result with units.

**04** Step 3: explain it in one sentence.

**FORMULA**

$$H_0: \beta_1 = 0$$

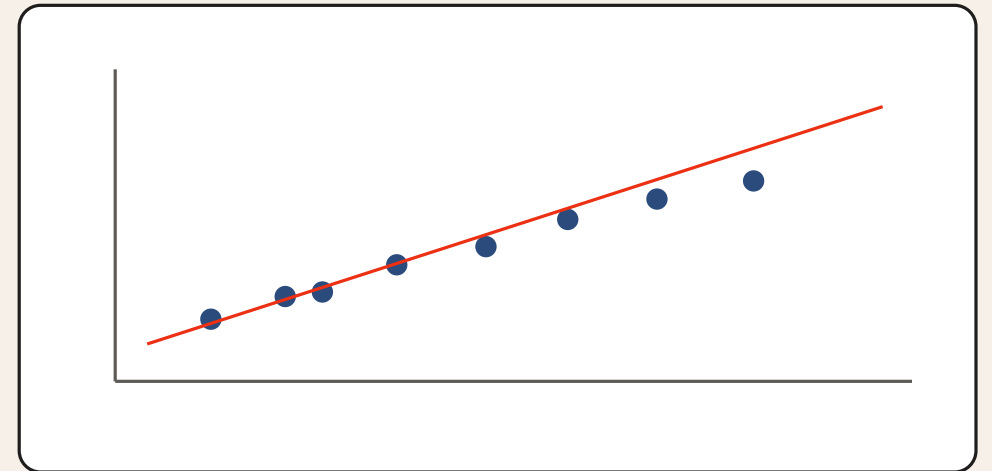
$$t = (b_1 - 0) / SE(b_1)$$

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## Visual intuition

- A graph should answer a question, not decorate the slide.
- Always label axes and units.
- Look for center, spread, shape and unusual values.

**FORMULA**

$$H_0: \beta_1 = 0$$

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## Second formula box

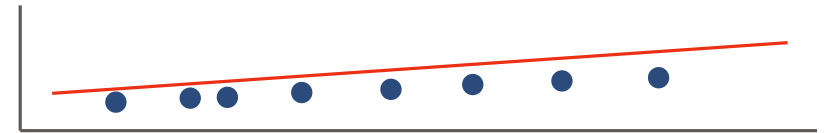
- This formula is separated from explanations but kept in one clean box.
- No broken equation fragments or floating text boxes.
- Notation is consistent across English and Russian versions.

**FORMULA 2**

$\text{Corr}(X,Y) \neq \text{causal effect}$

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + u$$

omit  $Z \rightarrow$  possible bias in  $\beta_1$



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## Common mistakes

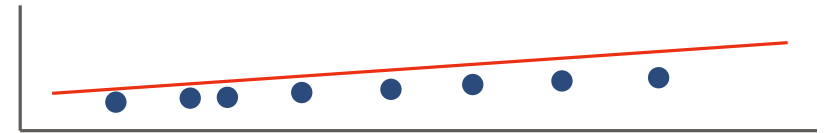
- Mixing sample and population symbols.
- Using a formula without checking conditions.
- Reporting a number without units or interpretation.

**FORMULA**

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## R / spreadsheet check

- Use software to calculate, but keep the statistical logic visible.
- Check variable names, missing values and units first.
- Reproducible work is easier to grade and correct.

**R / EXCEL**

```
# example pattern  
mean(x); sd(x); summary(x)  
plot(x, y); cor(x, y)
```

**FORMULA**

$$H_0: \beta_1 = 0$$

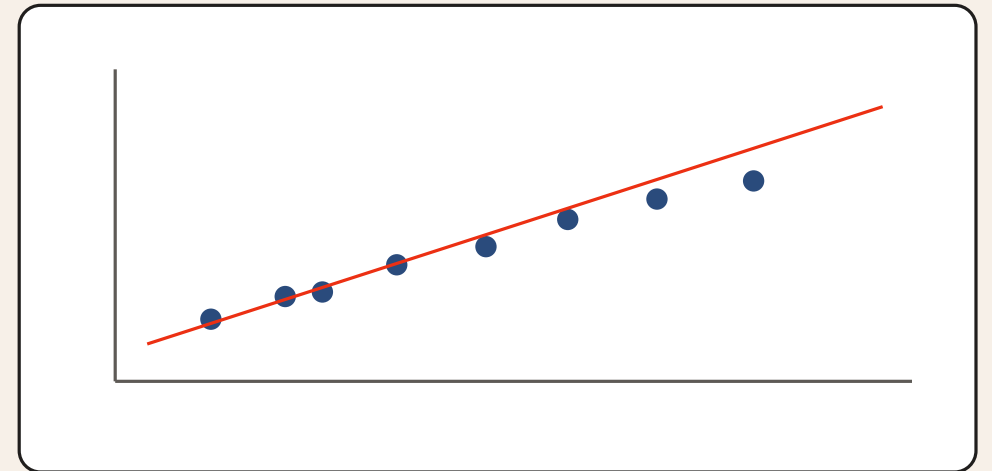
$$t = (b_1 - 0) / SE(b_1)$$

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## Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.

**FORMULA**

$$H_0: \beta_1 = 0$$

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## Conditions and assumptions

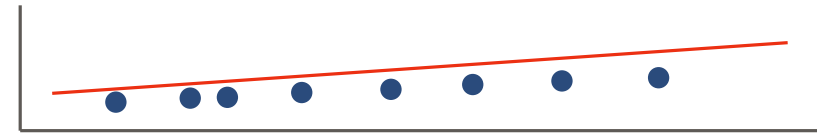
- Check whether the method fits the data type.
- Check independence and sample size when needed.
- If assumptions fail, explain the limitation.

**FORMULA**

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## Class exercise

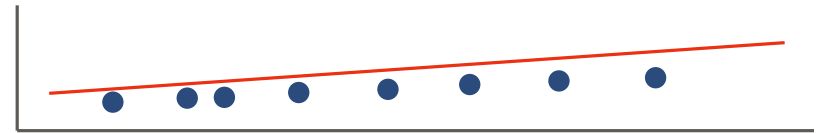
- Solve one calculation by hand.
- Then verify it with software.
- Write a two-sentence conclusion.

**FORMULA**

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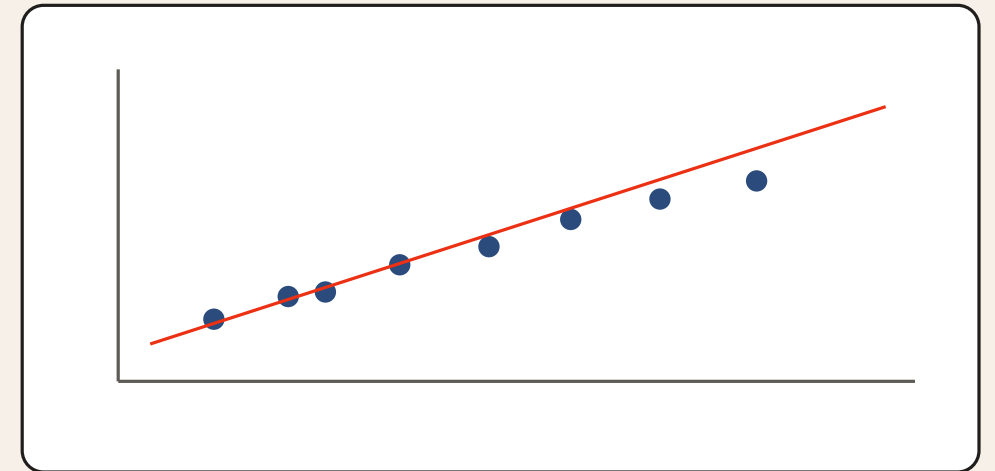


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## How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
- Separate statistical evidence from causal claims.

**FORMULA**

$$H_0: \beta_1 = 0$$

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## Final checklist

- Question is clear.
- Variables and units are defined.
- Formula is correct.
- Result is interpreted.
- Limitations are stated.

**FORMULA**

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